

Ankit Kumar

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Profile

CS student specializing in distributed systems & performance optimization. Built servers handling 80K+ RPS with custom data structures, protocols, and scalable architecture.

Skills

Java, Spring Boot, JavaScript/TypeScript, Express.js, Rust, Axum, Tokio, NIO, PostgreSQL, MongoDB, Redis, DragonflyDB, Docker, REST APIs, WebSockets, Distributed Systems, Performance Optimization, Concurrency, Benchmarking, Git

Experience

Backend Engineer Intern, ControlShift Talent Pvt. Ltd.

02/2025 – 05/2025

- Implemented WebSocket architecture for real-time features
- Optimized PostgreSQL indexes → reduced latency by **40%**
- Built async job system with BullMQ & Redis → improved throughput **3x**
- Developed **REST APIs** with validation & error handling

Projects

CinemaCore, Java 25, Spring Boot, PostgreSQL, Stripe · github.com/ankitkr-04/cinematicore

- Built cinema booking system achieving zero double-bookings via pessimistic locking & ACID transactions
- Engineered seat reservation with SELECT FOR UPDATE locking → handles 50+ concurrent bookings safely
- Integrated Stripe webhooks with idempotency checks → automated payment confirmation & refunds
- Implemented hold timer with async jobs → 5-min auto-release, email notifications

RedisCore, Java 24, NIO, Maven · github.com/ankitkr-04/redis-core

- Built Redis-compatible server achieving 80K+ RPS with distributed replication & sub-ms sync
- Implemented **25+ Redis** commands across 5 data types with full protocol compatibility
- Engineered **master-replica** replication system with automatic sync & sub-ms latency
- Developed thread-safe custom data structures (SkipList, QuickList)

Hyperlinkr, Rust, Axum, DragonflyDB, Docker · github.com/ankitkr-04/hyperlinkr

- Built high-performance URL shortener achieving **50K RPS** with caching & rate limiting
- Designed 4-layer caching (Bloom filters + Redis) → **95%+ hit rate**
- Optimized BASE62 encoder to **458K ops/sec** via concurrency & memory tuning
- Exposed REST APIs with analytics, device detection, rate limiting & error handling

Education

Bachelor of Technology in Computer Science,

2023 – Present

Lakshmi Narain College of Technology, Bhopal-462021

CGPA: 8.25

Certificates

Oracle Cloud Infrastructure 2025 Developer Professional (1Z0-1084-25) — Oracle |

Cisco Networking Essentials — Cisco Networking Academy | **Cloud Foundations** — AWS Academy